One-Proportion Test

2026-04-17

Introduction

The one-proportion test compares an observed success proportion $\hat{p}$ to a benchmark p_0. Applications: a phase II response rate compared to a historical control; a screening positive rate compared to a national reference; the success rate of a surgical procedure compared to a target.

Prerequisites

Binomial distribution, confidence intervals.

Theory

For $X \sim \text{Binomial}(n, p)$ with observed $\hat{p} = x/n$, three tests coexist:

Exact binomial test: computes the exact two-sided p-value under $H_0 : p = p_0$ using the binomial PMF. This is R's `binom.test()`. Always valid; no approximation.

Normal-approximation (Wald) z-test:

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} \sim \mathcal{N}(0, 1).$$

Works when $np_0 \geq 10$ and $n(1 - p_0) \geq 10$; biased and unreliable near $p = 0$ or $p = 1$.

Score (Wilson) test: uses the variance under H_0 and inverts to get a CI with better coverage than Wald, especially for small n or extreme p. R's `prop.test()` uses this by default.

Confidence intervals:

- Wald: $\hat{p} \pm 1.96\sqrt{\hat{p}(1 - \hat{p})/n}$ (poor near 0 or 1).
- Wilson: centred, always inside $[0, 1]$, good coverage.
- Clopper-Pearson (exact): always valid, conservative.

Assumptions

- Independent Bernoulli trials.
- Fixed n (not a sequential stopping rule).

R Implementation

```r
library(binom)

x <- 32; n <- 80; p0 <- 0.30

# Exact binomial test
binom.test(x, n, p = p0)
```

```r
# Score (Wilson) test
prop.test(x, n, p = p0, correct = FALSE)

# Normal-approximation test (manual)
phat <- x / n
z <- (phat - p0) / sqrt(p0 * (1 - p0) / n)
pval <- 2 * (1 - pnorm(abs(z)))
c(z = z, p = pval)

# Confidence intervals
binom.confint(x, n, conf.level = 0.95,
              methods = c("wald", "wilson", "exact"))
```

Output & Results

```
        Exact binomial test
x = 32, n = 80, p-value = 0.04215, 95% CI: (0.292, 0.517)

     1-sample test for given proportion without continuity correction
X-squared = 3.9286, df = 1, p-value = 0.04748, 95% CI (score): (0.303, 0.508)

     method  x  n    mean    lower   upper
1      wald 32 80   0.400   0.2926  0.5074
2    wilson 32 80   0.400   0.3020  0.5066
3     exact 32 80   0.400   0.2923  0.5172
```

Observed proportion 0.40 (95 % Wilson CI 0.30 to 0.51); exact binomial p = 0.042.

Interpretation

Report both the point estimate and the CI. In a manuscript: "the response rate was 40.0 % (32/80; 95 % Wilson CI 30.2 % to 50.7 %; exact binomial p = 0.042 vs. the null rate of 30 %)."

Practical Tips

- Use Wilson or Clopper-Pearson CIs by default; Wald is obsolete.
- For small n and extreme p (near 0 or 1), exact tests and intervals are mandatory.
- Do not report zero/one proportions as 0 and 1 with a Wald CI of "0 to 0"; use a Clopper-Pearson upper bound (rule of three: upper bound $\approx 3/n$ if $x = 0$).
- Continuity correction in `prop.test()` is unnecessary for modern purposes; set `correct = FALSE`.
- For agreement with two-proportion tests, use the same CI method (Wilson) in both.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem

- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests